



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - ATSB-style final report

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INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

At 02:47 on 12/08/2026, a 220 t class rigid haul truck, HT-407, was descending empty on the Stage 4 main ramp at Bindarra Gold Mine when a light vehicle, LV-118, drove out of the RL 312 dewatering spur into its path. The truck operator sighted it at about 40 m and braked and steered right towards the inner windrow within 1.4 seconds. The truck struck the light vehicle's right side at 24 km/h, pushing it 18.4 m down the ramp and onto its left side against the windrow.

The driver, a grade control technician employed by a survey subcontractor and on his third night shift at the mine, sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre at 07:40. His seatbelt was fastened, and the roll-over protection structure deformed without being breached.

The light vehicle was destroyed. The haul truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours.

What the investigation found

The spur was opened under a works order classified as routine earthworks, so no management of change was raised and no referral reached the traffic management plan owner, the survey department or the contractor's technical services group. The intersection was therefore absent from the traffic management plan, the in-cab road network and the collision awareness system configuration. It met the ramp 34 m beyond a crest, giving a descending truck 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m at the posted speed. It had no sign, give-way control or delineation.

Positive radio calls were required only at nine listed intersections, reissued annually with no means of adding a new road between reviews. Neither operator called. A suppression geofence extended in June 2026 covered the intersection, and it was likely that long silence on that ramp had led the truck operator to treat the absence of an alert as an empty road.

The light vehicle driver's site pit driving assessment, which the site standard said could not be substituted, had never been verified. He entered the ramp without stopping or establishing the truck's position. Both vehicles very likely arrived with neither operator aware of the other. On a finer balance, it was likely that the regulatory guideline also contributed by leaving the review trigger undefined and the interaction zones to the site, although the mine was not in breach.

Night shift emergency response relied on one paramedic, with the second responder called from a village 11 km away, delaying extrication by approximately 9 minutes. This, and the truck operator's roster exposure on night 10 of 14, did not contribute but increased risk.

Whether dust affected the truck operator's view could not be determined. The drivers' accounts differed, nothing at the site measures visibility, and the sighting distance matched the clear-air geometry. The truck operator's braking and steering reduced the impact speed from 37 km/h and kept the collision away from a drop of about 100 m.

What has been done as a result

At 05:10 on 12/08/2026, the Mining Manager directed a positive call before entering any Stage 4 intersection until further notice. The spur was barricaded and closed that night. On 14/08/2026,

survey measured sight distance at the ten other unlisted roads cut since the last review, and two below the plan minimum were closed pending a stop control. On 28/08/2026, Warrigal Mining Services began reporting suppressed collision awareness events. The investigation recommended changes to how new intersections reach BIN-TMP-004, to the management of change exclusion for earthworks, and to the control of geofence changes. Responses are due by 30/09/2026.

Safety message

A new road cut into an operating pit creates an intersection, whatever work authorised it, and every control that depends on knowing where intersections are needs to be told. Where a collision awareness system is suppressed, operators should know where and why, so that silence is never read as an empty road.

The occurrence

Before the night shift

The RL 312 dewatering spur was a 240 m access road cut in the Stage 4 open pit at Bindarra Gold Mine under works order WO-26-4471 to reach the new dewatering bore, DW-14. Earthworks were recorded complete on 31/07/2026. The bore was energised and light vehicles first used the spur on 02/08/2026, the day the works log recorded removal of the single "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth. Nothing replaced it. At the time of the occurrence the intersection had no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed for the spur.

The spur joined the main ramp from the east, on the inside of a left-hand bend, 34 m beyond the crest of the ramp. The survey department found that a descending operator at an eye height of 5.80 m first had line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first had line of sight to a descending truck at 38 m.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a descending haul truck, then reversed back and let it pass. At 23:48 he wrote hazard report HAZ-2026-0917, stating that "you cannot see up the ramp from that road", that "There is no sign on the ramp telling the trucks the road is there and there is no give way on our side", and that "I do not think they know the road is there." He rated the risk as Medium. On 06/08/2026 the Dewatering Superintendent closed the report with a Low rating, recording that the dewatering crew had been reminded at the pre-start on 05/08 "to take care entering the main ramp and to give way to haul trucks at all times." The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner.

The light vehicle driver was a grade control technician employed by survey subcontractor Merrivale Survey & Geotechnical. He completed Merrivale's two-hour online general mining induction on 24/07/2026. On 09/08/2026 he was issued a Bindarra site access card at the gatehouse under Merrivale's recognition agreement, after a 20 minute briefing on muster points, speed limits and personal protective equipment. There was no record of the Bindarra site-specific supplement or the Bindarra light vehicle pit driving assessment. He worked night shifts on 09/08, 10/08 and 11/08. He stated that on his first night the dewatering contractor's foreman had told him the spur was open and driven him out to it, and that he had since driven it six or seven times without meeting a haul truck at the intersection.

The night shift and the approach

The haul truck operator was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, with 6 hours 30 minutes of sleep in the preceding day. He had 14 months at Bindarra and 7 years on 220 t class rigid haul trucks. He described himself as tired in the way usual for night 10, but not falling asleep and driving properly.

At 02:22:41 on 12/08/2026, dispatch called the Stage 4 crews back after crib. HT-407 loaded at 02:31:18 and reported dumped at 02:38:55. At 02:46:18 dispatch told the excavator operator that HT-407 was about four minutes away. The truck was descending the main ramp empty on its second cycle after crib. The fleet management system recorded 41 km/h at RL 368 at 02:45:58 and 39 km/h at RL 348 at 02:46:31, with the retarder in the second position.

The truck's in-cab road network, last published on 24/07/2026, did not contain the spur and showed no intersection at RL 312. The truck operator said he knew the spur had been cut, believed it was still closed, and had noticed the closed sign was gone without thinking anything of it. The

night shift supervisor did not know a light vehicle was in the Stage 4 pit, and there was no permit, call-in or board by which the shift could have known.

At RL 312 the ramp ran approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, with a dry, well maintained surface. The driver finished levels at the DW-14 collar and drove the 240 m spur towards the ramp. From the spur length and his position at first sighting, the investigation estimated that he left the collar at about 02:45 and reached the spur mouth at about 02:47. His own estimate, about ten to three, was later than the recorded collision time and was not used.

The driver stated that he slowed at the spur mouth, looked left up the ramp, saw nothing and drove out at an estimated 25 km/h. On read-back of his interview notes he corrected his first account that he had stopped, saying he did not believe he had come to a complete stop. He described the view up the ramp as a piece of road and then nothing, because of the windrow and the rise, and said he had expected to see a truck's lights on the pit wall first. His radio was on channel 12, as the dewatering foreman had told him, and he did not recall hearing anything on it. The pit channel recording contained no call from either vehicle between the crib break and the collision. RL 312 was not one of the nine intersections at which the traffic management plan required a positive call.

The accounts of visibility differed. The truck operator stated there was some dust hanging but could not say whether it made a difference. The driver stated that the air was clear, that he could see the ramp surface and the windrow, and that there was no dust in the spur.

At 02:47:06 HT-407 recorded its last steady data point: 37 km/h at RL 314, 41 m above the intersection. The truck operator reported seeing the light vehicle at about 40 m, within a metre of the surveyed sight line. At 02:47:07.4, 1.4 seconds later, service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied. The truck operator said he braked and steered right, into the windrow, in one movement, because the left side was a drop of about 100 m with the light vehicle in front of him.

At 02:47:09.2 the collision awareness system on HT-407 logged the light vehicle's tag at 21.4 m, closing at 13 km/h, with the truck at 32 km/h. This was 1.8 seconds after braking began and 2.8 seconds before the collision. The event fell inside the S4-LOAD suppression geofence and no alert was presented in the cab.

The collision

At 02:47:12.0 the truck, at 24 km/h, struck the right side of LV-118, 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m from the inner windrow toe. Mean deceleration from brake application to impact was 1.85 m/s², against 1.9 m/s² in the brake test on 03/08/2026. Without braking, the collision speed would have been 37 km/h. LV-118 was pushed 18.4 m down the ramp, rolling onto its left side at 11.2 m where tyre marks changed to body panel gouging, and came to rest against the inner windrow facing down-ramp. HT-407 was stationary 19 m below the intersection at 02:47:14.3.

The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened and its webbing showed loading marks. The beacon whip was sheared at its base, and its lamp worked when tested after recovery. Damage to HT-407 was confined to the front left bumper, left mudguard and lower access ladder steps.

The response

The truck operator reported the collision on the pit channel at 02:47:41. At 02:48:01 dispatch stopped all Stage 4 units and told him to stay in the cab. He was cleared to get out at 02:48:44 and

reported at 02:49:30 that the driver was conscious and talking but could not get out. The supervisor directed that he not be moved.

The emergency response team was paged at 02:48. Its first vehicle entered the pit at 02:51 and reached the scene at 02:54. The night shift was rostered with one mines rescue paramedic. The second responder, required before any extrication, came from the accommodation village 11 km from the gatehouse and arrived at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. The post-incident review found that waiting for the second responder and the tool delayed extrication by approximately 9 minutes, and that this did not affect the clinical outcome because the injuries were stable and the patient was conscious throughout. The driver had a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was flown to the metropolitan trauma centre at 07:40.

The truck operator and the shift supervisor both tested negative for drugs and alcohol. At 05:10 the Mining Manager directed that a positive call on the pit channel was required before entering any intersection in the Stage 4 pit until further notice.

Context

Personnel information

Haul truck operator

The haul truck operator had 11 years in open pit mining, including 7 years on 220 t class rigid haul trucks across three sites. He had been employed by Warrigal Mining Services for 4 years 2 months and had worked at Bindarra for 14 months. He held current verifications of competency on the 220 t class unit (issued 06/2025), dozers and water carts. He completed the Bindarra site general induction in 06/2025 and his annual refresher in 02/2026.

His only Bindarra traffic management training was the general induction module, a 23 slide package naming the nine intersections at Appendix B of the traffic management plan. It did not cover unlisted intersections, and he had completed no traffic management refresher since. No person on site had training documentation covering the collision awareness system's suppression geofences.

He worked a 14 night on, 7 off roster, from 18:00 to 06:30 local, a 12 hour 30 minute shift including a 30 minute crib. Site access records confirmed his previous nine shifts were worked in sequence, none extended beyond 12 hours 40 minutes. The contractor's limits of 14 consecutive night shifts and a minimum 10 hour break were both met. His voluntary sleep diary showed between 5 and 7 hours of sleep on each of the nine days before the day preceding the occurrence.

Light vehicle driver

The light vehicle driver had 11 years in survey, 6 of them in mining, and held a surveying associate diploma. He had worked as a grade control technician with Merrivale Survey & Geotechnical since August 2022, rotating from the regional office between four client sites on a 9 on, 5 off roster, working 18:00 to 06:00. No duty or rest concerns were raised or identified. Bindarra was new to him, and 11/08/2026 was his third night shift there.

Neither Merrivale's training file nor the gatehouse access card register showed that he had been issued, completed or signed for the Bindarra site-specific supplement, or completed the Bindarra light vehicle pit driving assessment. He had never seen the Bindarra traffic management plan, BIN-TMP-004, and did not know it existed.

Night shift supervisor

The night shift supervisor had 19 years in open pit mining, 8 with Warrigal Mining Services, and had supervised the Stage 4 night shift for 2 years 6 months. He held current verifications of competency on haul trucks, excavators and dozers, completed his latest supervisor refresher in 11/2025, and worked the same roster as his crews.

He was responsible for the haul fleet, excavators, ancillary fleet, dozers and the dispatch operator, but not the dewatering crew, the survey contractor or any other light vehicle in the pit.

He knew the RL 312 spur had been cut and was used by the dewatering crew, but had not regarded it as an intersection. In about the middle of 2025 he learned from the technical services group that collision awareness alerts were suppressed around the loading face. He did not know the suppressed area had been extended in June 2026 to cover the ramp above the spur. His pre-start notes for the ten days after the spur opened did not mention it.

Vehicle and systems information

HT-407

HT-407 was a 220 t class rigid haul truck commissioned in 03/2021, with 31,480 engine hours. Its last 500 hour service was on 28/07/2026 with no defects carried forward. The 11/08/2026 pre-start check sheet was complete with no defects, and no defect had been raised against its lighting, mirrors, cameras or cab glass in the preceding twelve months. Brake performance tests on 03/08/2026 and 14/08/2026 found the retarder, service brakes and park brake within specification, with mean decelerations of 1.9 m/s² and 1.92 m/s² empty on a 9 per cent grade.

LV-118

LV-118 was a 4WD utility in service since 02/2024, with 61,200 km, last serviced on 17/07/2026. It had a certified roll-over protection structure, a 4.5 m amber beacon whip, reflective marking to the site standard and a UHF radio. The inspection sheet for the week of 09/08/2026 recorded all items serviceable. LV-118 was not dispatched, carried no FMS, and had a passive collision awareness tag.

Fleet management system road network

The FMS in-cab map for the Stage 4 pit was last published on 24/07/2026, before the spur was cut. Publishing a revised road network is a survey task raised by a management of change record.

Collision awareness system

The Stage 4 haul fleet's collision awareness system operated at "Level 7 (operator advisory)". "Level 9 (automated intervention)" was available but not enabled at Bindarra. The configuration was that "Advisory alerts are raised on a vehicle tag entering the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence." An alert was an advisory tone and a cab screen indication.

The S4-LOAD suppression geofence was drawn on 14/03/2025 around the Stage 4 loading face, where nuisance alerts among the excavator, trucks and service vehicles had made the system unusable. The technical services group extended it northward on 09/06/2026 as a configuration task when the face advanced, placing its boundary 180 m up-ramp of the RL 312 intersection. The extension was not recorded in, or reviewed against, the traffic management plan.

In the 12 months to 11/08/2026, HT-407 raised 412 advisory alerts, all outside the geofence. Across the Stage 4 haul fleet, 68 proximity events inside the geofence were logged and suppressed, nine involving a light vehicle tag on the main ramp below the boundary. Suppressed events were retrievable by query, but the technical services group's standing report covered only alert counts and system health.

The intersection and the conditions

The works order specified the spur as 240 m long and 8 m wide, windrowed both sides, meeting the ramp at an included angle of 71 degrees. The 1.6 m inner windrow beside the last 25 m of the spur's approach hid the lower body of a descending truck. The ramp's posted speed was 40 km/h.

The Stage 4 pit had no fixed lighting. The only lighting was vehicle lighting and the loading face light plants, 1.1 km away and out of sight behind the crest. Moonset was at 23:18 the previous evening.

The Bindarra met station at the ROM pad recorded south-east wind at 8 km/h, 9 degrees and no precipitation. The last rainfall was 26 days before the occurrence. The investigator examined the ramp at 06:20 and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection.

The water cart last watered the main ramp between RL 280 and RL 400 at 01:10, 1 hour 37 minutes before the occurrence. The standing instruction was a pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval, against a shift mean of 88 and a site trigger of 250. Nothing at Bindarra measures visibility.

Organisational and regulatory information

Regulatory guideline

Mines Safety Guideline MSG-14, revision 4, was issued by the State Mines Inspectorate in 2021 under regulation 6.22 of the Mining Operations Safety Regulations 2018. Compliance is one way of discharging the regulation, which requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine".

Clause 3.2 requires plan review at intervals not exceeding twelve months "and following any significant change to the mine's road network, vehicle fleet or operating pattern", without defining a significant change. Clause 5.1 requires nomination of interaction zones requiring positive communication, with no minimum or basis for nomination. Clause 5.4 allows light vehicle detectability by "line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these", without specifying which or how many. Under clause 3.5, the plan "is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."

On the day, Bindarra's plan met the guideline on its face: reviewed within twelve months, nominating nine interaction zones, and providing detectability by line of sight and elevated marking.

Bindarra traffic management plan

BIN-TMP-004 revision 7, owned by the Mining Manager, Corella Resources, was issued on 12/11/2025 with next review due 12/11/2026. Clause 4.1 requires that "Light vehicles shall give way to heavy vehicles at all times and in all circumstances." Clause 4.3.1 requires a positive call before entering any Appendix B intersection, and clause 4.3.2 provides that at other intersections "No call is required."

Clause 6.2 requires sight distance at every intersection of not less than the largest vehicle's stopping distance at the posted speed "and in no case less than 120 m", otherwise a stop, signalised or traffic controller control. The calculated stopping distance of a 220 t class truck empty at 40 km/h on a 9 per cent descending grade, using manufacturer's data and a 1.5 second response, is 96 m. Clause 6.4 requires warning signs, a give-way control and delineation only at intersections on the Schedule 1 road layout drawing.

Appendix B and Schedule 1 are reissued at the annual review, and nothing allows an intersection to be added between reviews or requires a new road to be referred to the plan's owner. Annual reviews were completed on 14/11/2023, 10/11/2024 and 12/11/2025. No out-of-cycle review under clause 9.1 had been held in the plan's eleven year history. The Stage 4 crib room board displayed the 12/11/2025 issue of Appendix B.

The survey department's road register recorded 14 new roads cut in Stage 4 since the 12/11/2025 review. The three cut under a management of change record were referred to the plan's owner. The 11 cut under a works order were not.

Management of change

Procedure CRL-PRO-022 clause 5.2 requires a management of change record before "a change to the mine's road network". Clause 5.3 excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced." Referrals to the plan owner, survey department and technical services group exist only in the management of change record, at clause 5.4. The works order form has no equivalent fields.

Works order WO-26-4471 was raised by the Mining Engineer on 28/07/2026 and approved by the Mining Superintendent on 29/07/2026. It classified the spur as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3." Its handover field read "N/A".

Contractor management

Contractor Management Standard CRL-STD-009 clause 7.3 allows a contractor's general induction to be accepted as equivalent where the contractor holds a recognition agreement, which Merrivale had held since 2023. Clause 7.4 excludes the site-specific supplement and pit driving assessment, which "shall not be substituted". Clause 7.5 makes the engaging department responsible for verifying 7.4 before a card is issued. The gatehouse issued cards against the recognition agreement alone and held no record of either item. The access card system did not distinguish people permitted to drive in an operating pit, and there was no permit, radio call-in, board or entry check for a light vehicle entering one.

Related occurrences

A search of Bindarra reports in the 24 months to 11/08/2026 returned four relevant hazard reports: HAZ-2026-0917 and three raised in 03/2025, 09/2025 and 02/2026. Each described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations.

All four were closed within three days by the receiving department, each recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the plan owner, and none changed Appendix B, Schedule 1 or any physical control.

Safety analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14 revision 4 was issued by the State Mines Inspectorate in 2021 under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". Compliance with the guideline is one way of meeting that requirement. The text of both documents was not in dispute, and it is virtually certain that the guideline left the following matters to the operator.

The guideline left three terms for the site to define:

- Clause 3.2 requires the traffic management plan to be reviewed "following any significant change to the mine's road network". The guideline does not define "significant change" and gives no example of one.
- Clause 5.1 requires the operator to nominate the intersections and interaction zones at which positive communication is required. It does not say who nominates them or on what basis, and it sets no minimum.
- Clause 5.4 provides that detectability of a light vehicle "may be provided by line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these". It does not say which of these, or how many, must be provided.

Clause 3.5 provides that the plan "is a site document" and that it is not approved by, and need not be submitted to, the Inspectorate except on request following an occurrence or an inspection.

On the day of the occurrence, Bindarra's plan met the guideline on its face. BIN-TMP-004 had been reviewed within twelve months. It nominated nine interaction zones and provided for detectability by line of sight and elevated marking. The mine was not in breach of the guideline or the regulation. This factor is therefore not a compliance matter, and it does not imply any failure by the Inspectorate. It concerns the width of the discretion the guideline leaves and how that discretion was taken up at this site.

At Bindarra, all three discretions were exercised through the same document, reviewed once a year, with no external review. Under the plan, a positive call is required only at an intersection listed at Appendix B. An intersection is added to Appendix B "where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication". Both the drawing and the appendix are reissued at the annual review. No provision in BIN-TMP-004 allows an intersection to be added between annual reviews. No provision requires a new road to be referred to the plan's owner when it is cut.

The document register showed annual reviews completed in November 2023, 2024 and 2025. No out-of-cycle review had been carried out in the plan's eleven year history. Since the November 2025 review, 14 new roads had been cut in the Stage 4 pit. The 3 roads cut under a management of change record were referred to the plan's owner. None of the 11 roads cut under a works order was referred. A search of the operator's reporting system over the preceding 24 months returned four hazard reports, including HAZ-2026-0917, describing a light vehicle and a haul truck meeting unexpectedly at intersections not listed at Appendix B. Each report was closed within three days with a reminder to the crew. None was referred to the plan's owner, and none led to any change to the plan or to any physical control.

The case that this factor contributed rests on that pattern. Where the external requirement leaves the review trigger undefined and the nomination of interaction zones open, there is no outside reference point against which a new road must be tested. A defined trigger and a stated minimum would have given the eleven unreferred roads, the RL 312 spur among them, somewhere to land. The internal controls that failed in this occurrence each failed for want of exactly such a reference point.

The investigation also considered evidence against that conclusion. The guideline did not prevent any action. Bindarra could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not stop a mine from listing every intersection it has. On this view, the mine did not fail to act because the guideline was open. It failed to act because nothing inside the mine put the question in front of the plan's owner. The external clause would then be background, and the organisational failure would be the whole explanation.

The investigation weighed these arguments and concluded that the breadth of the guideline's discretion likely contributed, by shaping the internal arrangements that followed. This conclusion is less certain than the other contributing findings in this report, and it is not presented as settled. The opposing reading, that the internal arrangements fully explain the outcome, could not be excluded.

The internal arrangements are findings in their own right and are argued in the analysis of organisational influences and risk controls. The spur was cut and opened to traffic under a works order with no management of change raised, which very likely contributed. Positive radio calls were required only at listed intersections, with the list rewritten once a year, which also very likely contributed. The finding here sits upstream of both. It does not displace them.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

The mine operator's Management of Change procedure, CRL-PRO-022, required at clause 5.2 that "a management of change record shall be raised before any of the following: a change to the mine's road network". Clause 5.3 then excluded "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced". Works order WO-26-4471 cut 240 m of access road to dewatering bore DW-14 at RL 312. It classified that work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3". The procedure, the works order and the survey department's road register were all documentary and agreed with each other. It is therefore virtually certain that the spur was cut and opened to traffic with no management of change record.

The consequence of that classification lay in what clause 5.4 attached to the record. Where a record was raised, the change owner was required to refer the change to "the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system configuration". Those three referrals existed only inside the management of change record. The works order form at CRL-PRO-022 Schedule 4 had no equivalent fields and no referral of any kind. WO-26-4471 carried no scope for signage, survey pickup, referral to the traffic management plan, publication of a revised road network or review of the collision awareness system geofences. Its handover field, the only place any of these could have been recorded, read "N/A".

None of the three referrals was made, and each of the three systems they served was left as it had been before the spur existed. The fleet management system in-cab map for the Stage 4 pit had

last been published on 24/07/2026, a week before the spur was completed. It still showed the main ramp with no intersection at RL 312 on the night of the occurrence. The intersection was not on the traffic management plan's road layout drawing. No review was made of the S4-LOAD suppression geofence, inside which the intersection lay.

A management of change record was the only route by which the new intersection could have reached the traffic management plan, the in-cab map or the geofence configuration. On that basis it is very likely that its absence contributed to the occurrence. The investigation considered the argument against this. A referral is not the same as a control, and the plan's owner, having been told of the spur, might still have considered it and left it off the list of intersections requiring a positive call. That possibility is why the influence is put at very likely rather than higher. It does not remove the effect on the in-cab map, which was a survey task raised only by a management of change record.

The arrangement was not confined to this spur. Of the 14 new roads cut in the Stage 4 pit since the 12/11/2025 review of the traffic management plan, three were raised under a management of change record and all three were referred to the plan's owner. The other 11 were raised under a works order and none was referred.

This factor led to two conditions that are argued elsewhere in this analysis. The first is the limitation of positive radio calls to the listed intersections, with that list rewritten once a year, which is discussed under Risk Controls. The second is the spur meeting the ramp beyond a crest with 41 m of sight line against a plan minimum of 120 m, which is discussed under Local Conditions.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

The Contractor Management Standard, CRL-STD-009, allowed a contractor's own general induction to be accepted as equivalent to the Bindarra general induction where the contractor held a current recognition agreement. Merrivale Survey & Geotechnical had held one since 2023. The standard drew a clear limit at clause 7.4. Equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted". Clause 7.5 made "the engaging department ... responsible for verifying that 7.4 has been completed before a site access card is issued".

The light vehicle driver's training file recorded a two hour Merrivale online induction completed on 24/07/2026 and a 20 minute gatehouse briefing on 09/08/2026, which covered muster points, speed limits and personal protective equipment. There was no record of the site-specific supplement being issued, completed or signed for, and no record of the pit driving assessment. The driver had never seen the traffic management plan, BIN-TMP-004, and did not know it existed. The gatehouse issued access cards against the recognition agreement alone. It held no record of either item and was not asked for one, and nothing in the access card system distinguished a person who could drive in an operating pit from one who could not. Merrivale's own records of the site supplement were not obtained, and the point was not put to the driver. For that reason it is extremely likely, rather than certain, that he drove in the operating pit without the supplement or the assessment.

The standard itself was sound and stated the requirement plainly. What did not work was its verification. The only point at which verification could have occurred was the issue of the access card, and the card system had no field for it.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice, including giving way, positive calls and what to do at a blind intersection. It is likely that

the absence of that assessment contributed to the light vehicle entering the ramp from the spur without stopping and without establishing the truck's position. That factor is argued under Individual Actions. Two matters weigh against a stronger conclusion. The rule the driver would have been taught, that light vehicles give way at all times, was one he was already applying. Separately, the sight line at the intersection was short enough to defeat a driver's attention without any shortfall in training.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system fitted to HT-407 operated at Level 7, operator advisory. Level 9, automated intervention, was available on the fleet and was not enabled at Bindarra. Advisory alerts were raised when a vehicle tag entered the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. The S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face to stop nuisance alerts. It was extended northward on 09/06/2026 when the face advanced, and its new boundary lay 180 m up-ramp of the RL 312 intersection. The contractor's technical services group made the extension as a system configuration task. It was not recorded in the traffic management plan and was not reviewed against it. The configuration, the event log and the twelve-month statistics came from the system itself and were downloaded sealed. It is virtually certain that the intersection lay inside a suppressed area.

The log recorded a proximity event at 02:47:09.2, with tag LV-118 at "range 21.4 m closing 13 km/h" and "alert SUPPRESSED (geofence)". Nothing was presented in the cab. Outside the geofence the same event would have raised an advisory tone and a screen indication.

The evidence on influence pointed two ways, and both readings are set out here. Against influence, the alert would have come 1.8 seconds after the operator had already begun braking and 2.8 seconds before impact. On this occasion, an advisory tone would have told him what he already knew. The system had no intervention capability, and with 41 m of sight line the outcome was fixed before either the alert or the sighting. On that reading, the suppression was a serious matter for the fleet but not a cause of this collision.

The investigation accepted a different mechanism. In the 12 months before the occurrence, HT-407 raised 412 advisory alerts, all of them outside the geofence. Across the Stage 4 fleet, 68 proximity events inside the geofence were suppressed, nine of them involving a light vehicle tag on the main ramp below the boundary. That is the same arrangement of vehicles and location as the occurrence. These events were visible to anyone who queried the system, but nobody did. The technical services group's standing report covered alert counts and system health and did not report suppressed events at all. No training record on site, for any person, covered the geofences.

Both the truck operator and the supervisor raised the suppression without being prompted. The operator's account indicated that a silent system on that ramp had come to mean an empty road. That expectation was a condition he carried into the crest. On that basis, it was likely that the suppression contributed to neither operator being aware of the other as the vehicles reached the intersection. The concern was not that a geofence existed, since suppression around an excavator is needed to keep the system usable. The concern was that its boundary moved 180 m in June 2026 and nobody outside the group that moved it was told.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

BIN-TMP-004 clause 4.3.1 required that "a positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B". Clause 4.3.2 provided that at other intersections "No call is required." Appendix B listed nine intersections. Its note provided that an intersection is added "where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication". Both Appendix B and Schedule 1 were reissued only at the annual review. No provision allowed an intersection to be added between reviews, and no provision required a new road to be referred to the plan's owner when it was cut. The RL 312 spur appeared on neither document. These were documentary matters, and it is virtually certain that the plan's call requirement did not apply at this intersection.

The pattern extended beyond this spur. Since the 12/11/2025 review, 14 new roads had been cut in the Stage 4 pit. The 11 cut under a works order were not referred to the plan's owner, and no out-of-cycle review had been held in the plan's eleven-year history. Three earlier hazard reports, as well as HAZ-2026-0917, described light vehicles and haul trucks meeting unexpectedly at intersections not listed at Appendix B. Each was closed with a reminder to the crew, and none changed the appendix, the drawing or any physical control. The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine intersections on the crib room board and nowhere else.

The pit channel extract contains no call by either vehicle about the spur, the ramp or a light vehicle before HT-407 reported the collision at 02:47:41. Had a call been required, the channel was open and both vehicles were on it. The dispatcher was also tracking the truck closely enough to place it "about four minutes off" the loading face at 02:46:18. A call made but not acknowledged would still have left a driver deciding whether to proceed. Even so, it was very likely that the limitation of the call requirement to listed intersections contributed to neither operator making a positive call. The absence of those calls is argued under Individual Actions. The upstream conditions, the guidance that left the review trigger undefined and the spur opened without a management of change record, are argued under External Influences and Organisational Influences.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift was rostered with one qualified mines rescue paramedic at the surface medical facility. A second responder, required before any extrication, was called in from the accommodation village 11 km from the gatehouse. The roster, the response log and the post-incident review agreed on the times, so it is virtually certain that this arrangement existed. Dispatch paged the team at 02:48. The first response vehicle reached the scene at 02:54, and the second responder arrived with the hydraulic rescue tool at 03:07. The driver was extricated at 03:14. The post-incident review found the extrication was delayed by approximately 9 minutes and that the delay did not affect the clinical outcome.

The collision and the driver's injuries were complete before the team was paged. It is therefore unlikely that the response arrangement influenced the occurrence or its outcome. It is nonetheless very likely that the arrangement increased risk. The driver was conscious, stable, restrained and protected by an intact roll-over structure. A fuel fire, an unconscious casualty or a compromised airway would have made the same 9 minutes significant. The site's night shift extrication capability depended on one person plus a drive from the village, and nothing in the arrangement changed with the severity of the event.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

The geometry of the intersection set the conditions within which every other factor in this occurrence operated. The spur met the main ramp 34 m beyond the high point of a crest. An operator descending at an eye height of 5.80 m first had line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first had line of sight to a descending truck at 38 m, and the 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach.

The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade was 96 m. This figure was calculated with the manufacturer's data and a 1.5 second response. Clause 6.2 of the traffic management plan required the sight distance at every intersection to be not less than that stopping distance "and in no case less than 120 m". Where that could not be achieved, it required a stop, signalised or controlled intersection. The survey concluded that the intersection "does not meet clause 6.2 on either limb."

The intersection had no warning sign on either ramp approach and no give-way or stop control. It also had no delineation and no posted speed for the spur. The construction sign at the mouth was removed on 02/08/2026 and nothing replaced it. Clause 6.4 would have required these controls, but it applied only to intersections shown on the plan's road layout drawing, and the RL 312 spur was not on it. Seven days before the occurrence, a dewatering technician described the same conditions from the same position: "you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything."

It is virtually certain that this sight line existed. It was surveyed to both eye heights and was independently described in the hazard report. At 37 km/h, 41 m gave the truck operator about four seconds, and the stopping distance was more than twice the sight line. A truck at the posted speed could not stop for anything at this intersection, however alert its operator. The drivers' attention, their training and any radio call all came after a geometry that could not be driven safely. The sight line was only hazardous once a vehicle was in it, and the factors that placed the light vehicle there are argued in other subsections. With that qualification, it is very likely that the restricted sight line contributed to both vehicles arriving at the intersection with neither operator aware of the other. The way the spur came to be opened without the plan's intersection controls is addressed under Organisational Influences.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

The investigation examined whether airborne dust reduced the truck operator's ability to see the light vehicle. It could not resolve the question.

Some evidence pointed towards dust:

- The ramp was last watered at 01:10, 1 hour 37 minutes before the occurrence, against a standing night shift interval of 90 minutes. A pass was therefore 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval. The shift mean was 88.
- The truck operator described dust hanging on the ramp after the cart's earlier pass.
- The truck descended behind its own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver.

Other evidence pointed against it:

- The light vehicle driver had been in the same place about four seconds earlier. He was firm that the air was clear and that he could see the ramp surface and the windrow.
- The truck operator declined to say that dust made a difference.
- The reading of 142 was well below the site trigger of 250. It measured particulate at a monitor head 180 m away, not what could be seen from either cab, and nothing at Bindarra measures visibility.
- Most significantly, the operator sighted the light vehicle at 41 m. The survey showed this was the distance the geometry allowed in clear air, which left dust with nothing to explain.

The overdue water cart pass and the elevated reading were real deviations at the relevant time and place. Their effect on detection was not established in either direction. The two drivers' accounts did not agree, and only a measurement of what could be seen from the cab on the night could have settled it. No such measurement exists. It was about as likely as not that dust reduced the operator's view. The matter remains unresolved, and nothing from it is carried into the findings.

The ramp surface was wet and the truck could not stop in the distance available

A wet surface was ruled out as very unlikely for four reasons:

- The met station recorded no precipitation, and there had been no rain for 26 days.
- The investigator found the ramp dry at 06:20.
- The braked tyre scuff showed no locked or sliding wheel.
- The deceleration achieved, 1.85 m/s², matched the unit's tested dry figure of 1.9 m/s².

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

The truck operator was on night 10 of a 14 night roster. The collision occurred at 02:47, 10 hours 47 minutes into a 12 hour 30 minute shift and within the period of lowest circadian alertness. Every limit in the contractor's fatigue management standard had been met. The operator's voluntary sleep record showed 6 hours 30 minutes of sleep the day before and between 5 and 7 hours on each of the preceding nine days. It is virtually certain that this exposure existed.

Detection in an unlit pit at this hour is the task fatigue degrades first. However, the recorded data showed a response of 1.4 seconds from the last steady data point to the first brake pressure rise. This was inside the 1.5 second response assumed in the plan's stopping distance calculation. The braking and steering that followed were immediate and in the only available direction. With a stopping distance more than twice the sight line, a faster response would not have provided the margin needed. There was no slowed response or omission in the data for fatigue to have worked through. It is unlikely that fatigue influenced the occurrence.

The exposure nevertheless remains a concern. The Stage 4 night roster places every haul truck operator at the controls of a 220 tonne vehicle in the small hours of a tenth consecutive night. The pit has no fixed lighting, so the operator's own eyes are the only defence against an unlit hazard at an unlisted intersection. The same roster is worked on three other pits on the site. It is likely that this increased risk.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

The pit channel was logged continuously, and the investigation listened to the whole period from the return from crib at 02:22:41 to 2 minutes after the collision. No transmission from either vehicle

referred to the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. Apart from logging on at the start of shift at 18:04, LV-118 made no transmission at all that shift. The truck operator, the light vehicle driver and the night shift supervisor each confirmed this in interview. It is virtually certain that no call was made.

No call was required. Clause 4.3.2 of BIN-TMP-004 states that at intersections not listed at Appendix B "No call is required", and the RL 312 spur intersection was not listed. The three interviewees independently described the same local practice. Calls were made at the nine intersections on the crib room board and nowhere else. The supervisor stated that he had never corrected anyone for not calling elsewhere, and that requiring calls everywhere would make the channel unusable. The omission therefore matched both the written procedure and the established practice on the pit. It was not a departure by either operator. The reasons the intersection was absent from Appendix B, and why the list was revised only once a year, are discussed under Risk Controls.

A call from either vehicle would have been heard by the other vehicle, by dispatch and by the supervisor. Dispatch was already following the truck closely enough to place it, telling the excavator operator at 02:46:18 that "407's about four minutes off you". A call was therefore very likely to have made each operator aware of the other. A call provides awareness, however, and not separation. The two vehicles would still have had to establish who went first across an intersection with 41 m of sight line. On that basis, the absence of a positive call likely contributed to both vehicles arriving at the intersection with neither operator aware of the other.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

The driver's account was that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out. When his account was read back to him, he corrected it from having stopped to having slowed, and said he did not believe he had come to a complete stop. That correction was against his own interest. The scene evidence was consistent with it. The impact position was 6.2 m into the ramp from the eastern edge of the spur mouth, and the light vehicle's gouge ran from that point. It is virtually certain that the light vehicle entered the ramp without stopping and without establishing the truck's position.

The driver described himself as applying the rule in clause 4.1 that light vehicles give way to heavy vehicles at all times. Nothing at the spur mouth prompted him to do more. There was no give-way control, no stop line, no warning sign and no posted speed, because clause 6.4 applied only to intersections on the road layout drawing. He had not completed the Bindarra light vehicle pit driving assessment and had never seen the traffic management plan. That gap in his preparation is analysed under Organisational Influences.

A full stop at the spur mouth would have changed little. The survey gave 38 m of sight line from the mouth at his eye height. The inner windrow hid the truck's lower body for the last 25 m of the spur's approach. A look to the left from a stationary position would therefore have shown him approximately what a look at 25 km/h showed him. Even from a stop, he would have gained a few metres of sight line and not the 96 m the truck needed to stop. Without the light vehicle on the ramp there would have been no collision, and entering without establishing the truck's position was the proximate step. On that basis, this action very likely contributed to both vehicles arriving at the intersection with neither operator aware of the other. It was taken in conditions that gave the driver no reliable means of establishing the truck's position.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

The fleet management system download recorded the truck's last steady data point at 02:47:06, at 37 km/h and 41 m above the intersection. Service brake pressure rose from 0 to 620 kPa in 0.4

seconds from 02:47:07.4. The retarder was taken to maximum and 14 degrees of right steering was applied in the same movement. The scene marks corroborated the download. This 1.4 second response was within the 1.5 seconds allowed in the stopping distance calculation used to assess the intersection against clause 6.2. Mean deceleration was 1.85 m/s², against 1.9 m/s² recorded in the unit's monthly brake test. The truck struck the light vehicle at 24 km/h. Without braking, the impact speed would have been 37 km/h.

The steering input was as significant as the braking. The braked scuff deviated 1.4 m to the right over its first 20 m, toward the wall side. That was the only direction that did not risk carrying the light vehicle over a 100 m drop. The light vehicle came to rest on its left side against the inner windrow. After stopping 19 m below the intersection, the operator reported the collision and remained in the cab as dispatch instructed until the ramp was stopped. He then went to the light vehicle and stayed with the driver until the emergency response team arrived.

The operator's response very likely reduced the severity of the outcome. Braking substantially reduced the impact speed and energy. The steering placed the collision against the wall side of the road.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

The sections on risk controls, local conditions and individual actions each set out a separate condition. These were the alerts suppressed inside the extended geofence, the 41 m sight line against a plan minimum of 120 m, and the absence of any positive call. They also include the light vehicle entering the ramp without stopping. Each was found to have contributed, and each is argued in its own subsection. They are treated here as parts of one mechanism, because each led to the same condition: two vehicles reached the RL 312 intersection at night with neither operator aware of the other until about four seconds remained.

On the truck's side, HT-407 was descending empty at 37 km/h. Its in-cab road network showed no intersection at RL 312. There was no warning sign on the approach and no call on the channel. The collision awareness system logged LV-118 at 21.4 m and raised nothing in the cab. On the light vehicle's side, LV-118 left the spur at about 25 km/h. The driver had no give-way control and made no call, and had no knowledge of the truck. The survey found that "A driver at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m". The descending operator first had line of sight to the spur mouth at 41 m.

The first sighting on the download matches that geometry. The truck's last steady data point was at 02:47:06, 41 m above the intersection. Service brake pressure began to rise 1.4 seconds later. The pit channel recording, the fleet management system download, the collision awareness log and all three interviews give no indication that either operator knew of the other before then. The night shift supervisor did not know a light vehicle was in the pit at all. There was no permit, call-in or board by which he could have known.

One limit applies to this conclusion. The light vehicle carried no fleet management system, and its CAS transponder was a passive tag. Its movements before the final seconds therefore rest on the driver's own account alone. The recorded data and the other interviews are consistent with that account, and nothing contradicts it. On that basis it was extremely likely that both vehicles arrived at the intersection with neither operator aware of the other.

It was also extremely likely that this condition led to the collision. The collision followed directly from the convergence. Once each vehicle first became visible to the other at about 41 m, nothing

remaining in the sequence could have prevented the impact. The truck operator's braking and steering reduced the impact speed but could not avoid the collision.

A further possibility was that dust on the ramp reduced the truck operator's ability to see the light vehicle. This was examined but could not be resolved, and it is discussed in the local conditions subsection. It forms no part of this conclusion.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The collision occurred at 02:47:12, 4.6 seconds after the brake application. The point of impact was 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the inner windrow toe. The truck's accelerometer logged 1.9 g longitudinal. Without braking, the impact speed would have been 37 km/h rather than 24 km/h.

The recorded data, the damage pattern and the debris field all agree on how the vehicles met. Damage to LV-118 was concentrated on the right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The geometry of the intersection explains why the light vehicle was struck on its side. The spur joins the ramp from the east, on the inside of a left-hand bend, at an included angle of 71 degrees. A vehicle emerging from it therefore presented its side to a truck descending the ramp. A 220 tonne truck striking the side of a light vehicle of about 3 tonnes leaves the outcome to the light vehicle's structure, its restraint and the impact speed.

It was virtually certain that the impact occurred as described. It was virtually certain that the impact was the means by which the convergence became an injury.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from a tyre scuff to a body panel gouge, which is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The door on that side was against the ground, so the driver was extricated through the rear of the vehicle.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. These are the injuries a side impact and roll-over produce when the structure holds and the restraint is worn, and both did so here. The roll-over protection structure was deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened, with loading marks on the webbing.

The scene measurements, the vehicle examination and the medical evidence agree. It was virtually certain that the roll-over occurred. It was virtually certain that the roll-over was how the collision produced the injuries that made this a serious occurrence.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

The possibility that the truck's service brakes were degraded was considered and found to be very unlikely, because mean deceleration of 1.9 m/s² on 03/08/2026 and 1.92 m/s² on 14/08/2026 in brake tests bracketed the 1.85 m/s² recorded during the occurrence, and the unit had no brake, retarder or park brake defect raised in the preceding twelve months.

Findings

From the evidence available, the following findings are made with respect to the collision between a haul truck and a light vehicle at the RL 312 intersection on the main ramp of the Stage 4 open pit at Bindarra Gold Mine on 12/08/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- It is virtually certain that collision awareness alerts on HT-407 were suppressed inside the S4-LOAD geofence, which had been extended on 09/06/2026 to a boundary 180 m up-ramp of the new intersection. It is likely that this contributed, not through the alert itself (which would have come 1.8 seconds after braking began and told the operator what he already knew), but because the absence of any alert inside the suppressed area had led the truck operator to treat silence as meaning an empty road. (Safety Issue)
- Mines Safety Guideline MSG-14 revision 4 left the review trigger undefined and left the nomination of interaction zones and the means of light vehicle detectability to the site, which is virtually certain, and it is likely, on a finer balance than the other contributing factors, that this width of discretion contributed by giving new roads no defined point at which they had to reach the traffic management plan, although the mine was not in breach and nothing in the guideline prevented an out-of-cycle review. (Safety Issue)
- The RL 312 access spur was cut and opened to traffic under works order WO-26-4471, classified as routine earthworks under CRL-PRO-022 clause 5.3 with no management of change raised, and this very likely contributed because the referrals to the traffic management plan owner, the survey department and the technical services group existed only within that record, which left the plan, the in-cab road network and the geofence configuration unchanged. (Safety Issue)
- With virtual certainty, the spur met the ramp 34 m beyond a crest, giving a descending truck operator 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m at the posted 40 km/h, and it is very likely that this contributed by leaving about four seconds between first sighting and the intersection, a distance in which a truck at the posted speed could not stop.
- BIN-TMP-004 virtually certainly required positive radio calls only at the nine intersections listed at Appendix B, a list rewritten once a year at the annual review with no provision for adding an intersection between reviews or for referring a new road to the plan's owner, and this very likely contributed because the RL 312 intersection was not listed and so no call was required there. (Safety Issue)
- It is virtually certain that neither operator made a positive call on the pit channel before entering the intersection, where the plan required none and local practice matched the plan, and this likely contributed, although a call would have given each operator knowledge of the other rather than separation, which still had to be resolved within 41 m of sight line.
- It is very likely that the contractor's general induction was accepted as equivalent and that the Bindarra site-specific supplement and light vehicle pit driving assessment were never verified, because the gatehouse issued access cards against the recognition agreement alone and the access card system had no field for either item. This likely contributed, although the give-way rule the assessment would have taught was one the

driver was already applying and the sight line limited what any training could achieve. (Safety Issue)

- On leaving the spur at about 25 km/h, the light vehicle driver entered the ramp without stopping and without establishing the truck's position, which is virtually certain, at an intersection with no give-way control, stop line, warning sign or posted speed, and this very likely contributed, although the survey showed that a full stop would have given him only a few metres more sight line and not the 96 m the truck needed.
- Both vehicles very likely arrived at the intersection with neither operator aware of the other, the first sighting available to either being at 41 m and about four seconds, and this very likely contributed by bringing them together at a point where nothing remained that could prevent a collision.
- At 02:47:12.0 the haul truck's front left struck the right side of the light vehicle at 24 km/h, which is virtually certain, and it is virtually certain that the impact contributed as the means by which the convergence of the two vehicles became an injury.
- It is virtually certain that the light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow, and virtually certain that this contributed to the driver's injuries, which were limited by a fastened seatbelt and a roll-over protection structure that deformed without being breached.

Other factors that increased risk

- It is virtually certain that the night shift was rostered with one emergency response paramedic, with the second responder required before any extrication called in from the accommodation village 11 km from the gatehouse, which delayed extrication by approximately 9 minutes. Although this did not influence the occurrence or the clinical outcome, it very likely increased risk because the night shift extrication capability did not change with the severity of the casualty. (Safety Issue)
- The truck operator was on night 10 of 14, in the eleventh hour of a 12.5 hour night shift and in the circadian low, with every limit in the contractor's fatigue management standard met, and while the recorded 1.4 second response and the braking and steering that followed did not support any effect of fatigue on this occasion, it is likely that this standing roster exposure increased risk for the haul truck operators who work it.

Other findings

- It is very unlikely that the ramp surface was wet and the truck could not stop in the distance available, as the ramp was found dry, no rain had fallen for 26 days and the achieved mean deceleration of 1.85 m/s² matched the unit's dry tested figure, and the hypothesis was examined and ruled out.
- Brake tests on 03/08/2026 and 14/08/2026 and the occurrence download showed it to be very unlikely that the haul truck's service brakes were degraded and could not achieve the required stopping performance, and that hypothesis was examined and ruled out.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting, which very likely reduced risk by lowering the impact speed from 37 km/h to 24 km/h and keeping the collision on the wall side of the road, away from a drop of about 100 m.

Safety issues and actions

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14 revision 4 required a site's traffic management plan to be reviewed after any significant change to the road network but did not define that term. It also left the operator to nominate its own interaction zones, with no minimum and no basis for the nomination, and allowed light vehicle detectability to be met by any one of five means. Under clause 3.5 the plan was a site document, not approved by the Inspectorate or submitted to it except on request after an occurrence. At Bindarra all three of these matters were settled in BIN-TMP-004, a document reviewed once a year with no outside review. The plan met the guideline on its face on the day of the occurrence, and the mine was not in breach.

The issue is the width of the discretion the guideline leaves to the site. It is not a failure by the Inspectorate. With no defined point at which a new road had to reach the plan, 11 of the 14 roads cut in Stage 4 since the last review were never referred to the plan's owner. The RL 312 spur was one of them. The investigation found that this likely contributed to the occurrence, but on a finer balance than the other contributing factors. Nothing in MSG-14 prevented an out-of-cycle review or the listing of every intersection, and the decisive gap was that nothing inside the mine put the question to the plan's owner.

No safety action has been recorded in response to this safety issue.

Access spur cut and opened to traffic under a works order, with no management of change raised

The RL 312 access spur was cut and opened to traffic under works order WO-26-4471, which classified it as routine earthworks within the approved Stage 4 design. As a result, no management of change record was raised for it under CRL-PRO-022.

Such a record would have triggered three referrals at clause 5.4:

- to the traffic management plan owner
- to the survey department, for the fleet management system road network
- to the contractor's technical services group, for the collision awareness system geofences.

None of these referrals was made. The plan, the in-cab road network and the geofence configuration were therefore left unchanged.

The significance of the issue extends beyond the RL 312 intersection. A works order built an intersection into a haul road, and the classification on its front page removed every consequence of having done so. Fourteen new roads have been cut in Stage 4 since the traffic management plan's review of 12/11/2025. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

On 01/09/2026 the investigator in charge recommended one of two changes. The first was that CRL-PRO-022 clause 5.3 stop excluding earthworks that create a new intersection with a haul road. The second was that the works order form at Schedule 4 carry the three referrals the management of change record holds at clause 5.4. The recommendation is open, with a response due by 30/09/2026.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

It is extremely likely that the light vehicle driver's general induction with his employer, Merrivale Survey & Geotechnical, was accepted as equivalent under the Contractor Management Standard CRL-STD-009, and that the Bindarra site-specific supplement and the Bindarra light vehicle pit driving assessment were never verified before he was issued a site access card. The standard itself was sound. Clause 7.4 excluded both items from equivalence and required the pit driving assessment to be completed at site by every person who would drive in an operating pit. Clause 7.5 made the engaging department responsible for verifying this before a card was issued.

The gatehouse issued cards against the recognition agreement alone. It held no record of either item and was not asked for one. The access card system did not distinguish a person who might drive in an operating pit from a person who might not, so there was no point at which the verification required by clause 7.5 could take place. The driver's training file recorded the Merrivale online module of 24/07/2026 and a 20 minute gatehouse briefing, with no supplement and no pit driving assessment. He had not seen BIN-TMP-004.

The safety significance is that a person could drive a light vehicle on an active haul road, in this case on his third night at the mine, without the assessment the site's standard said could not be substituted. No part of the system that admitted him was checking for it. This likely contributed to the occurrence.

No safety action has been recorded in response to this safety issue.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

It is virtually certain that the Bindarra traffic management plan, BIN-TMP-004, required a positive call on the pit channel only before entering one of the nine intersections listed at Appendix B, and required no call anywhere else. An intersection could reach that list only through the road layout drawing at Schedule 1. That drawing was reissued only at the annual review. The plan had no provision for adding an intersection between reviews and no requirement for a new road to be referred to the plan's owner when it was cut. The RL 312 spur appeared on neither document.

This was the mine's own defence against exactly this interaction, and at this intersection it could not apply under the plan's own terms. It is very likely that this contributed, because no call was required at the RL 312 intersection and none was made. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. Eleven of the fourteen new Stage 4 roads cut since the last review had never been referred, and no out-of-cycle review had been held in the plan's eleven year history.

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager, Corella Resources, issued a site-wide direction. It required a positive call on the pit channel before entering any intersection in the Stage 4 pit, listed or not, until further notice. The direction was read at the following four pre-starts and posted beside the Appendix B board in the Stage 4 crib room. It remains in force until BIN-TMP-004 is amended.

The investigation has also proposed that BIN-TMP-004 provide a route by which an intersection reaches Appendix B between annual reviews, and that the plan require a new road to be referred to its owner when it is opened to traffic.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

On 09/06/2026 the contractor's technical services group extended the S4-LOAD suppression geofence on the Stage 4 haul fleet's collision awareness system 180 m up-ramp of the RL 312 intersection. This placed the intersection inside an area where advisory alerts were suppressed, and the change was neither recorded in the traffic management plan nor reviewed against it. (Safety Issue)

The issue matters because the system was fitted, working and silent exactly where it was needed. It detected the light vehicle at 21.4 m and raised nothing. In the twelve months before the occurrence, HT-407 raised 412 advisory alerts, none of them inside the geofence. Over the same period, 68 proximity events inside it were suppressed, nine of them a light vehicle on the main ramp. The standing report covered only alert counts and system health, so these suppressed events were reported to nobody. No person on site had been trained on the geofences. Operators therefore had no way of knowing that silence on the ramp did not mean the road was clear.

Safety action by Warrigal Mining Services

On 28/08/2026, Warrigal technical services added suppressed proximity events to the monthly collision awareness report, broken down by geofence and by vehicle class. It also re-cut the previous twelve months for the Stage 4 fleet on the same basis. The action was in progress, with the first report due on 30/09/2026.

Proposed safety action

The investigation recommended that any change to a suppression geofence boundary be treated as a change to a documented control under CRL-PRO-022 clause 5.2. It also recommended that operators be told in writing where the collision awareness system is suppressed and why. A response was due by 30/09/2026.

One emergency response paramedic rostered on night shift, with the second responder called from the village

It is virtually certain that the Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who had to be present before any extrication could begin, was called in from the accommodation village 11 km from the gatehouse. The roster, the emergency response log and the post-incident review agreed on the times.

On the night of the occurrence, dispatch paged the team at 02:48. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived with the hydraulic rescue tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The site's post-incident review found that extrication had been delayed by approximately 9 minutes while the team waited for the second responder and the tool.

The collision, the injuries and their severity were all complete before the emergency response was paged. The post-incident review recorded, on clinical grounds, that the delay did not affect the outcome because the driver's injuries were stable and he was conscious throughout.

The arrangement nonetheless very likely increased risk. It was a structural gap in a last line of defence, present on every night shift. Nothing about it changed with the severity of the casualty. In a fuel fire, with an unconscious casualty, or where an airway required the vehicle to be opened within the first ten minutes, the same wait could have led to a different outcome.

No safety action has been recorded in response to this safety issue.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse